

NEUROCROWD

AI-Based Real-Time Crowd Intelligence and Risk Prediction System

AEGISVISION

01. Team Details

02. Problem Statement
and Objectives

04. Existing Solutions and
Identified Gaps

05. System Design

06. Implementation
Progress

07. Conclusion

AEGISVISION

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PROBLEM STATEMENT

Large public places such as railway stations, stadiums, festivals, and transport hubs are highly vulnerable to overcrowding and stampede risks. Existing CCTV systems are reactive—they only record incidents after they occur and rely on continuous human monitoring, delaying emergency response. NeuroCrowd transforms conventional surveillance into an AI-powered predictive safety system by analyzing real-time crowd density, movement patterns, and abnormal behavior to identify potential risks before they escalate. The system provides early risk scores and instant alerts, enabling authorities to take proactive action, improve crowd management, and enhance public safety using existing CCTV infrastructure.

EXISTING SOLUTIONS AND IDENTIFIED GAPS

Existing Solutions

1. Traditional CCTV Surveillance
2. Manual Crowd Monitoring
3. Basic People Counting Systems
4. Conventional Video Analytics

Identified Gaps

1. Records incidents after they occur without predicting potential risks.
2. Depends on continuous human observation, leading to delays and human errors.
3. Counts people but does not analyze crowd movement or abnormal behavior.
4. Provides limited insights and lacks real-time predictive decision support for authorities.

ARCHITECTURE

NeuroCrowd – High-Level System Architecture



IMPLEMENTATION PROGRESS

A working prototype of NeuroCrowd has been developed to process live CCTV/webcam video streams for person detection, crowd counting, density estimation, motion analysis, turbulence analysis, and AI-based risk prediction. An interactive Streamlit dashboard visualizes live person count, crowd density, motion level, turbulence index, risk percentage, and real-time alerts. The system successfully demonstrates end-to-end crowd monitoring and predictive analytics. Model accuracy and crowd counting performance are currently being optimized for dense and highly congested environments as part of ongoing development.



 Crowd Safety System

TOTAL PEOPLE: 10
RISK: SAFE
MOTION: 0.99 | TURB: 1.47
PANIC: NO

CONCLUSION

NeuroCrowd transforms conventional CCTV surveillance into an AI-powered predictive crowd intelligence system by enabling real-time crowd monitoring and early risk assessment. By analyzing crowd density, movement patterns, turbulence, and risk levels, the system supports authorities in making timely decisions to prevent overcrowding and potential emergencies. With further optimization and real-world deployment, NeuroCrowd has the potential to enhance public safety and contribute to smarter, safer urban environments.

THANK YOU